

Supplementary Information

A Calibrated Transcriptomic Atlas of Porcine Development
Reveals Conserved Molecular Programs with Humans

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Supplementary Figures

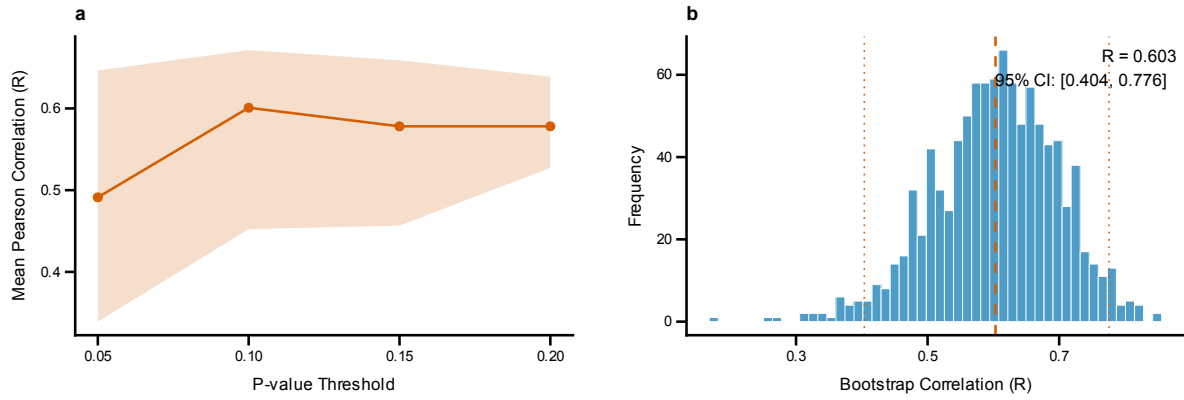


Figure 1. Cross-species sensitivity analysis. **a**, Mean Pearson correlation between pig and human developmental fold changes across different p-value thresholds. Error bars show min-max range across different numbers of genes tested. **b**, Bootstrap distribution of correlation coefficient (1,000 iterations) for the optimal parameter set ($p < 0.10$, $n = 36$ genes). Dashed line indicates mean correlation; dotted lines show 95% confidence interval.

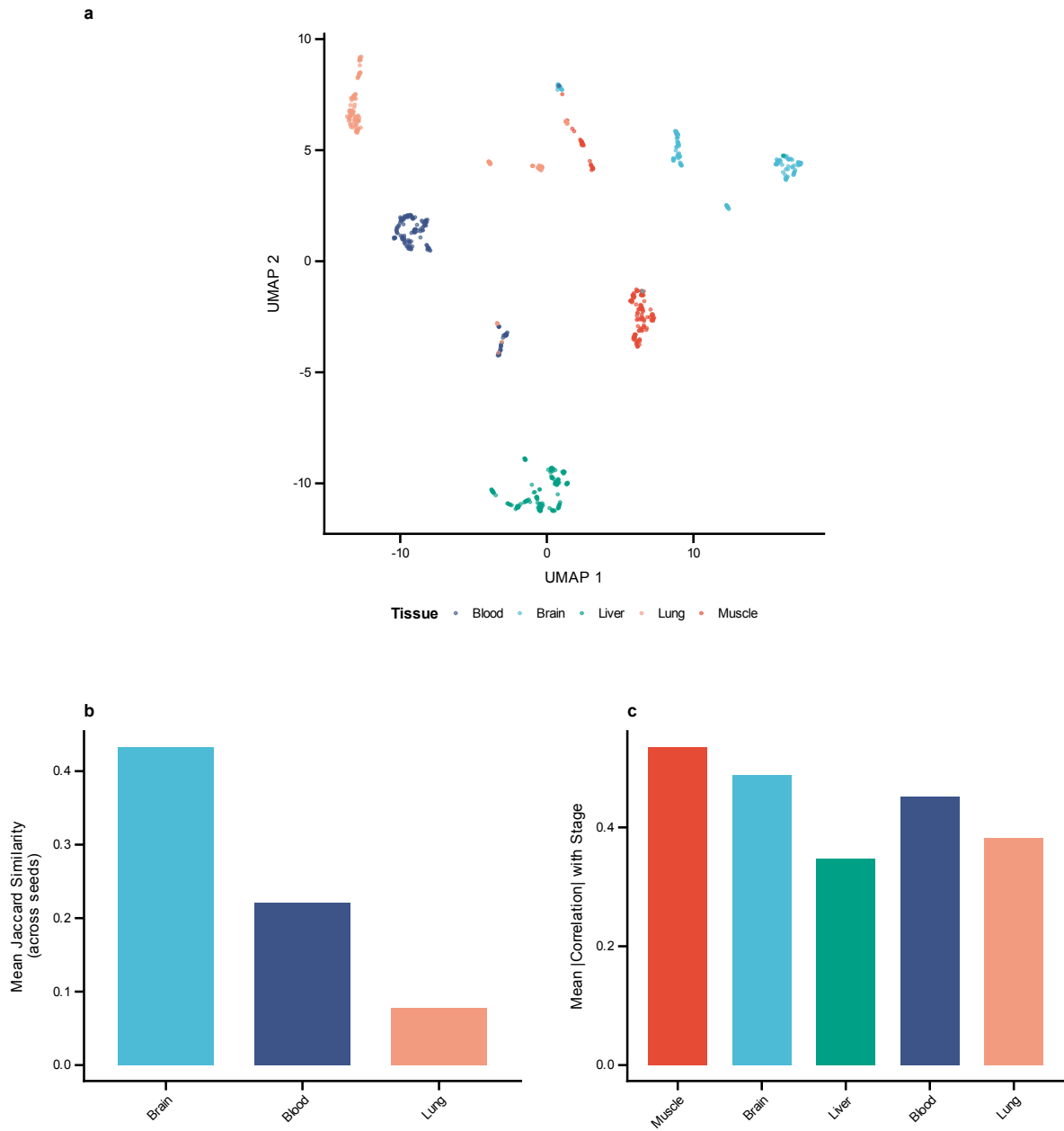


Figure 2. Tissue specificity and feature stability. **a**, UMAP visualization of expression profiles across five tissues, demonstrating clear tissue-specific clustering. **b**, Within-tissue feature stability measured as mean Jaccard similarity of top 50 features across 5 random seeds. Higher values indicate more stable feature selection. **c**, Mean absolute correlation of top features with developmental stage within each tissue, validating biological relevance of selected markers.

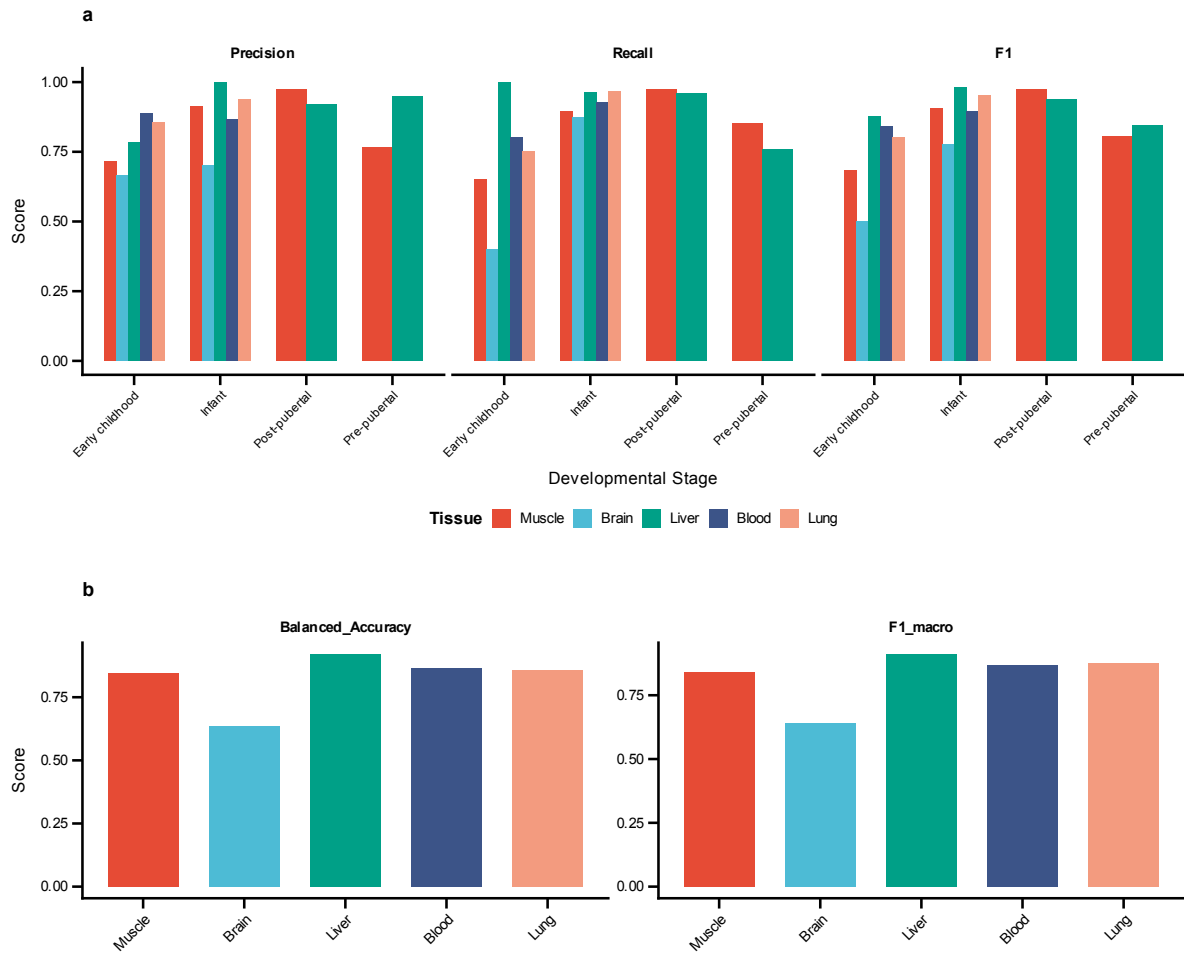


Figure 3. Extended performance metrics. **a**, Per-class precision, recall, and F1 scores for each developmental stage across all tissues. **b**, Overall balanced accuracy and F1-macro scores on test set for each tissue.

Supplementary Tables

Table 1. Train versus test performance comparison. Test set performance (30% held out) is reported for all tissues. Train set metrics are not shown as models were evaluated only on held-out data to prevent overfitting.

Tissue	Scheme	n	Balanced Acc.	F1-macro	F1-weighted	MAE
Muscle	4-class	761	0.844	0.842	0.908	0.10
Brain	2-class	43	0.637	0.639	0.671	–
Liver	4-class	309	0.920	0.910	0.913	0.09
Blood	2-class	78	0.864	0.869	0.874	–
Lung	2-class	129	0.859	0.876	0.921	–

Table 2. Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the number of test samples per class.

Tissue	Stage	n	Precision	Recall	F1
Muscle	Infant	58	0.912	0.897	0.904
Muscle	Early childhood	23	0.714	0.652	0.682
Muscle	Pre-pubertal	27	0.767	0.852	0.807
Muscle	Post-pubertal	121	0.975	0.975	0.975
Brain	Infant	8	0.700	0.875	0.778
Brain	Early childhood	5	0.667	0.400	0.500
Liver	Infant	26	1.000	0.962	0.980
Liver	Early childhood	18	0.783	1.000	0.878
Liver	Pre-pubertal	25	0.950	0.760	0.844
Liver	Post-pubertal	24	0.920	0.958	0.939
Blood	Infant	14	0.867	0.929	0.897
Blood	Early childhood	10	0.889	0.800	0.842
Lung	Infant	31	0.938	0.968	0.952
Lung	Early childhood	8	0.857	0.750	0.800

Table 3. Functional annotations of key developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. All listed genes show conserved developmental expression patterns between pig and human muscle tissue.

Gene	Function	Developmental Role
<i>SORCS2</i>	VPS10 receptor	Regulates motor neuron innervation and synaptic maturation during postnatal development
<i>FGFRL1</i>	FGF receptor-like 1	Muscle development and differentiation; may influence fibre-type balance
<i>KREMEN1</i>	Transmembrane protein 1	Modulates Wnt/ β -catenin signalling in neuromuscular development
<i>ACHE</i>	Acetylcholinesterase	Neuromuscular junction function; marker of synaptic maturity
<i>MSS51</i>	Translational activator	Negative regulator of oxidative metabolism; downregulated during growth
<i>IGSF1</i>	Ig superfamily member 1	Pituitary glycoprotein involved in hormone regulation
<i>PDLIM3</i>	PDZ/LIM domain 3	Muscle cytoskeletal protein at Z-discs; expression decreases postnatally